HK IMO Training 2020-2021
Problem Set 8
For 19 December 2020

Problem 29. There are $n \geqslant 4$ people in a meeting. Some pairs of people know each other. Suppose each person knows at least one other person but no one knows all other people. Prove that we can find 4 people and arrange them in a circle such that all of them knows exactly one of their two neighbours.

Problem 30. Let n be a positive integer. In a village, n boys and n girls are living.

For the yearly ball, n dancing couples need to be formed, each of which consists of one boy and one girl. Every girl submits a list, which consists of the name of the boy with whom she wants to dance the most, together with zero or more names of other boys with whom she wants to dance. It turns out that n dancing couples can be formed in such a way that every girl is paired with a boy who is on her list.

Show that it is possible to form n dancing couples in such a way that every girl is paired with a boy who is on her list, and at least one girl is paired with the boy with whom she wants to dance the most.

Problem 31. There are some bus routes in a city. Each bus route has at least four stops, and any two bus routes have exactly one stop in common. Prove that the stops can be partitioned into two groups such that each bus route includes stops from each group.

Problem 32. In a country, there are at least two cities. We can travel from some cities to some other cities by plane or by train (possibly both). Some routes may operate in a single direction only. For every pair of cities A and B, we can go from A to B, or from B to A (or both) by travelling on only one type of vehicles (possibly through some other cities). Prove that there exists a city such that any other city can be reached by travelling on only one type of vehicles. (It is possible to reach different cities using different types of vehicles.)